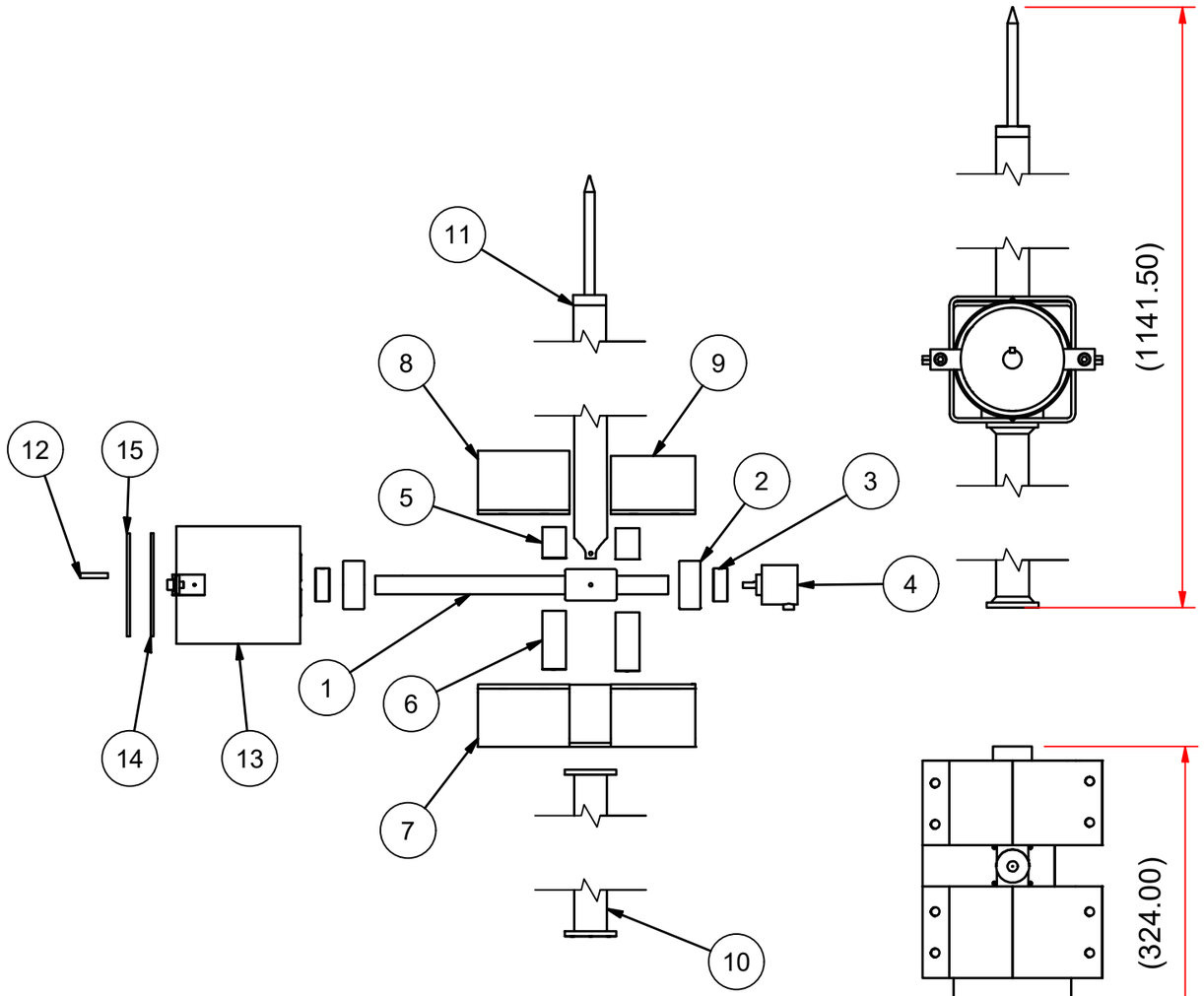
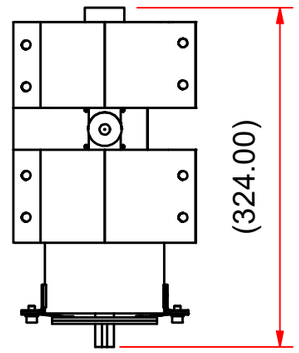


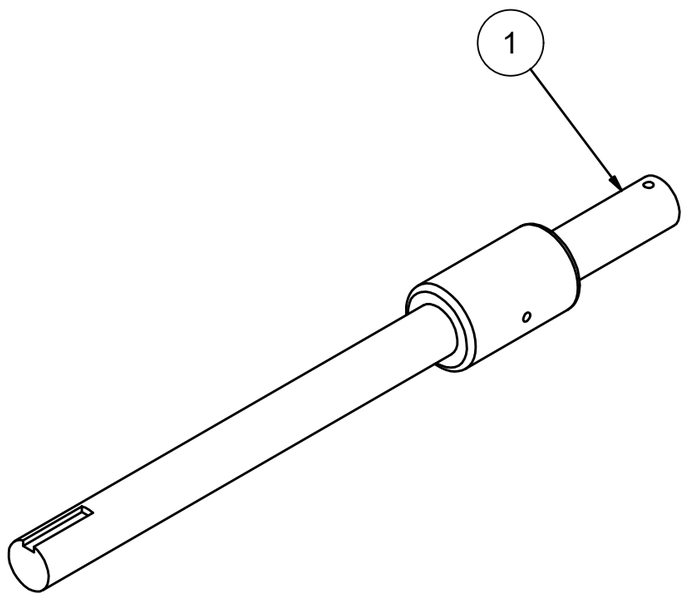


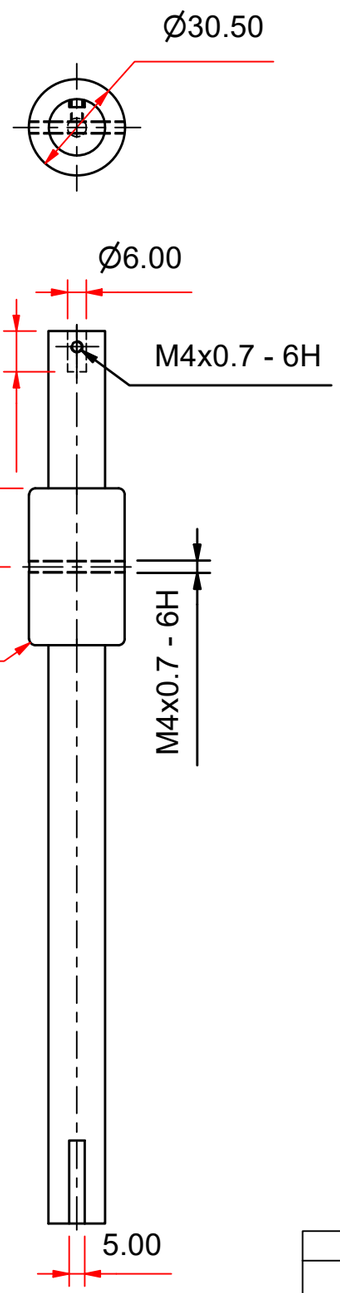
Parts List			
15	24989142-15-00-00 BRAKE DISC	1	WOOD
14	24989142-14-00-00 FRICTION PLATE	1	MILD STEEL
13	24989142-13-00-00 ELECTROMAGNET ASSEMBLY	1	CUSTOM - MILD STEEL
12	24989142-12-00-00 BRAKE DISC KEY	1	ALUMINIUM
11	24989142-11-00-00 ARTICULATING LINK	1	ALUMINIUM
10	24989142-10-00-00 BASE LINK	1	ALUMINIUM
9	24989142-09-00-00 JOINT HOUSING TOP-BACK	1	ALUMINIUM
8	24989142-08-00-00-00 JOINT HOUSING TOP-FRONT	1	ALUMINIUM
7	24989142-04-00-00 JOINT HOUSING BOTTOM	1	ALUMINIUM
6	24989142-03-00-00 BEARING HOUSING BOTTOM	2	ALUMINIUM
5	24989142-02-00-00 BEARING HOUSING TOP	2	ALUMINIUM
4	GTS06-OC-RA1000B-2M OPTICAL ENCODER 1000PPR	1	-
3	8236976 RS PRO SHAFT COLLAR	2	-
2	2204-2RSTN RS PRO 20 MM ID ROW BALL BEARING	2	-
1	24989142-01-00-00 JOINT SHAFT	1	ALUMINIUM
ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

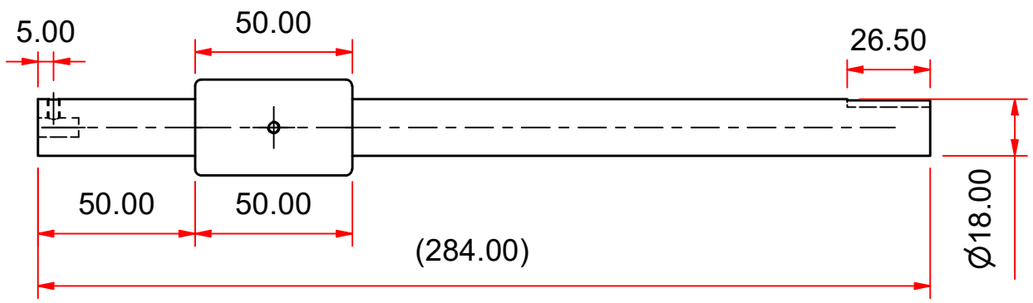
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

STELLENBOSCH UNIVERSITY

STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A**3 = 1:6 **UNITS IN** mm	**TITLE:** ELECTROMAGNETIC BRAKE JOINT	
			DATE 2026/10/01	**SHEET Nr.** 1 **OF** 30 **SHEETS**	**Nr.** 24989142-00-00-00







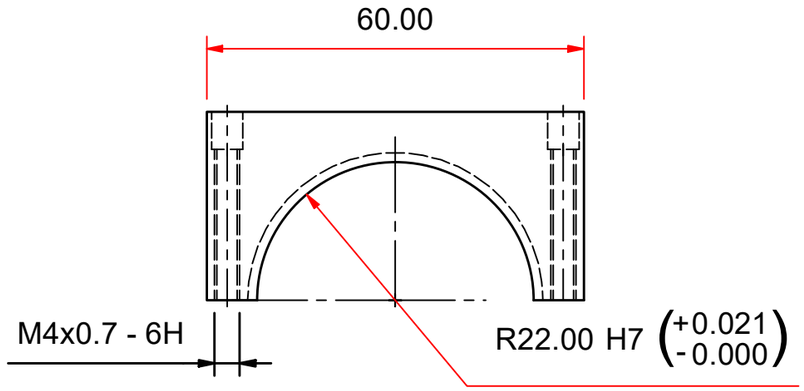
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

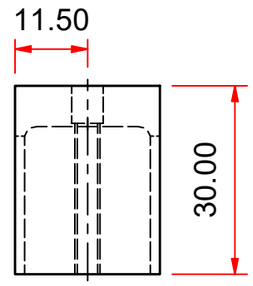
Parts List

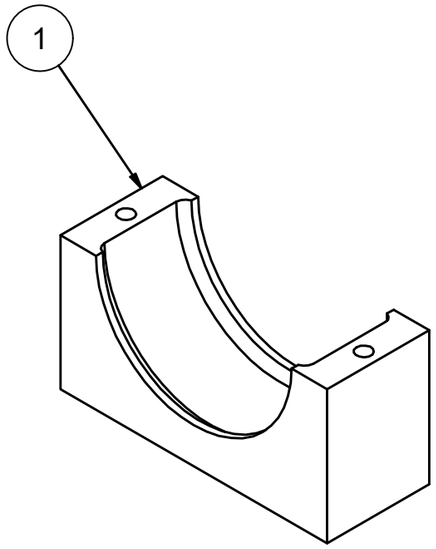
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
1	24989142-01-00-00 JOINT SHAFT	1	ALUMINIUM

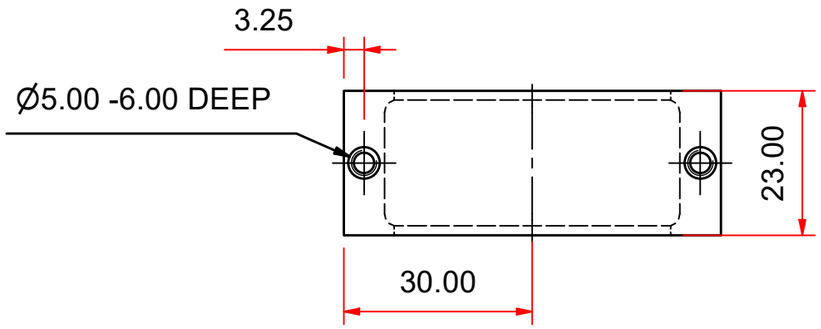
STELLENBOSCH UNIVERSITY

STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**DATE** 2026/10/01	**SCALE ON A3** = 1:2 **UNITS IN** mm	**TITLE:** JOINT SHAFT **SHEET Nr.** **OF** **SHEETS** **Nr.** 24989142-01-00-00









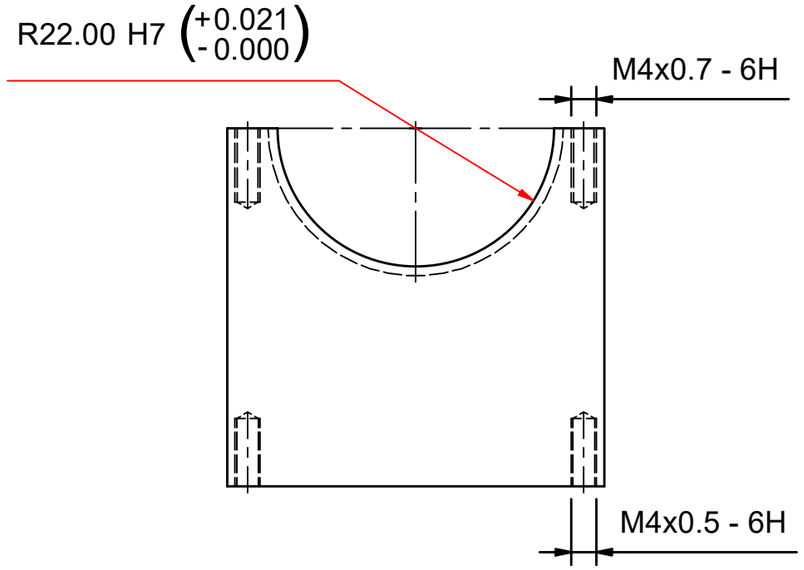
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

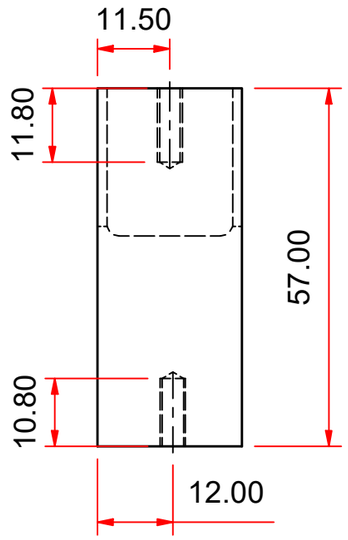
Parts List

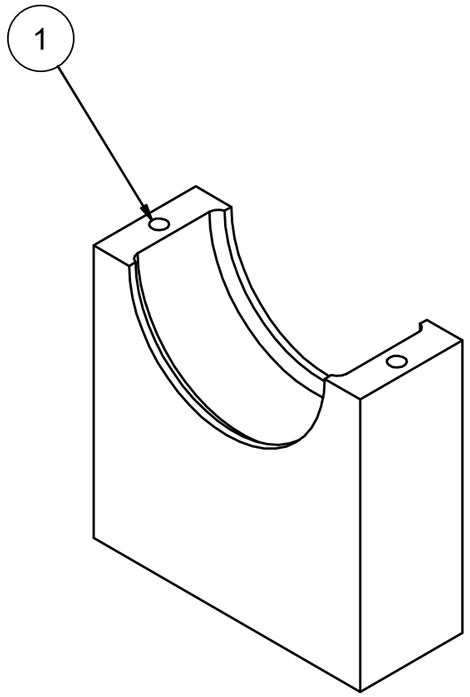
1	24989142-02-00-00 BEARING HOUSING TOP	2	ALUMINIUM
ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

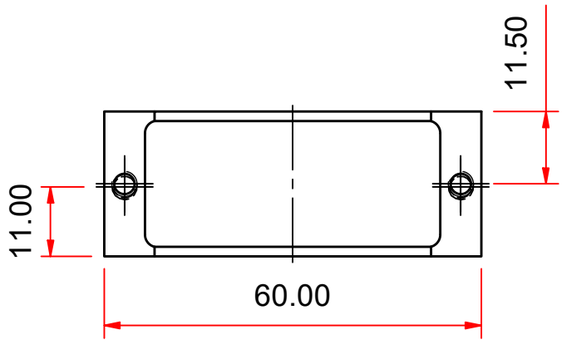
STELLENBOSCH UNIVERSITY

STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A3** = 1:1 **UNITS IN** mm	**TITLE:** BEARING HOUSING TOP	**Nr.** 24989142-02-00-00
			DATE 2026/10/01	**SHEET Nr.** **OF** **SHEETS**	









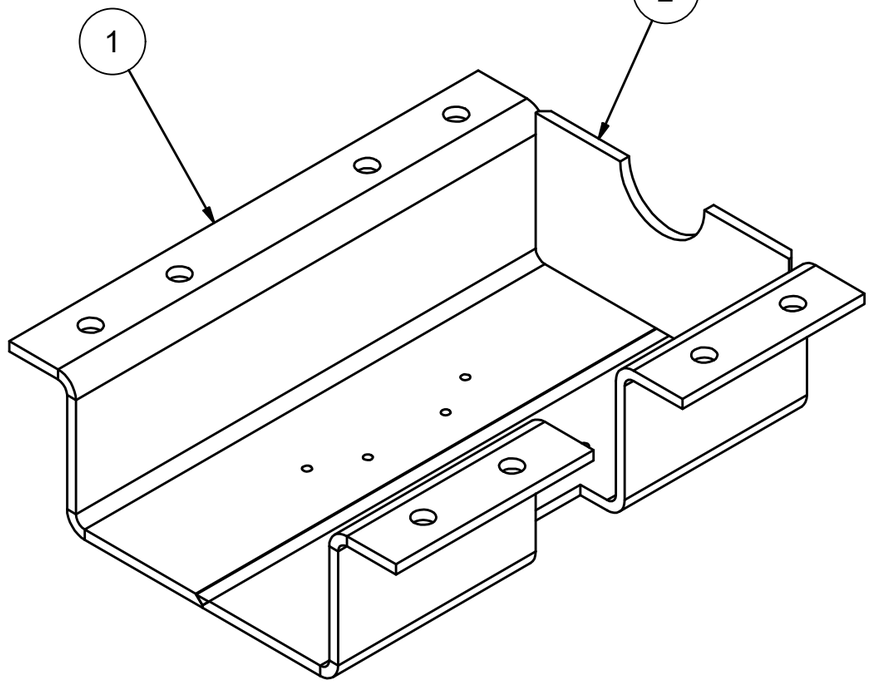
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

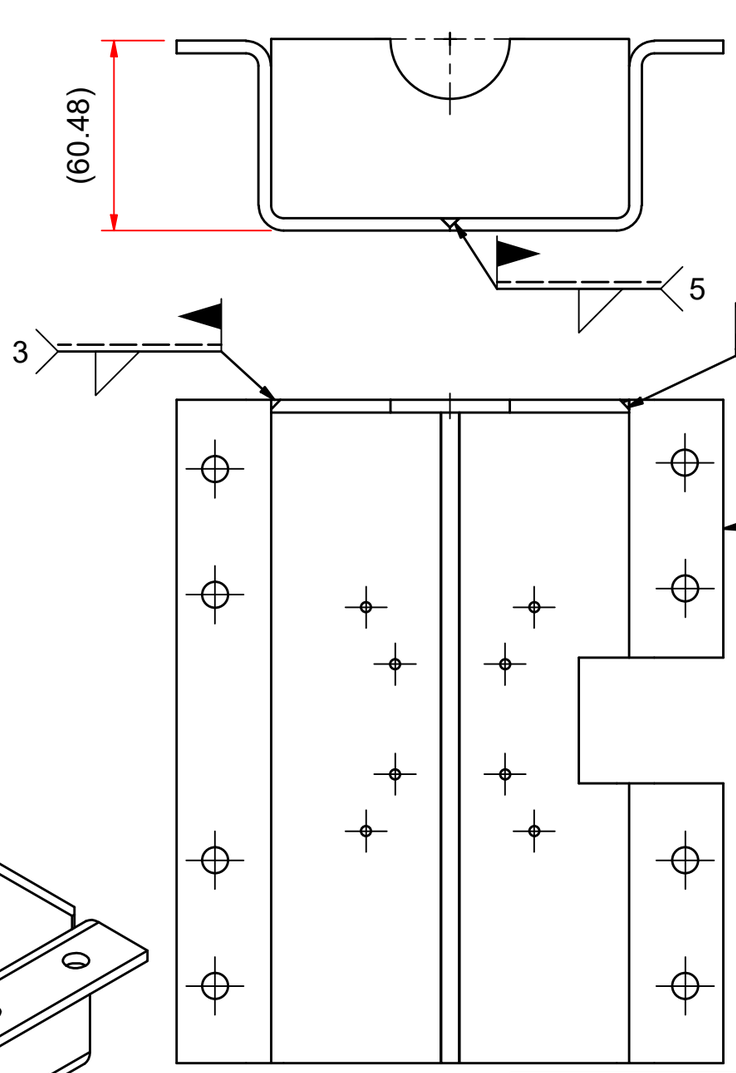
Parts List

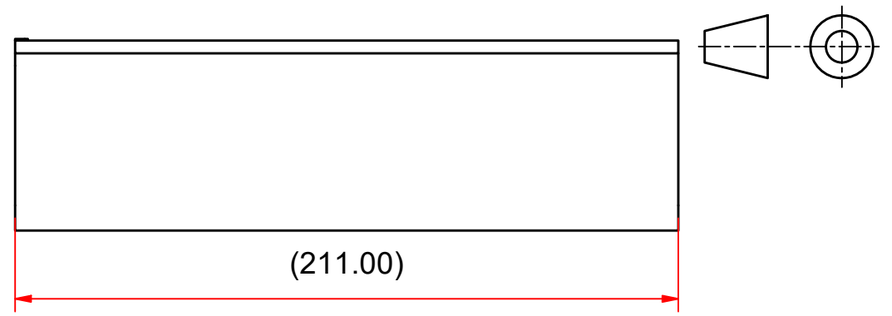
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
1	24989142-03-00-00 BEARING HOUSING BOTTOM	2	ALUMINIUM

STELLENBOSCH UNIVERSITY

STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A**3 = 1:1 **UNITS IN** mm	**TITLE:** BEARING HOUSING BOTTOM	
			DATE 2026/10/01	**SHEET Nr.** OF **SHEETS**	**Nr.** 24989142-03-00-00







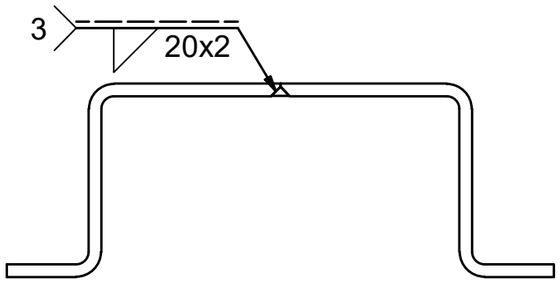
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

Parts List

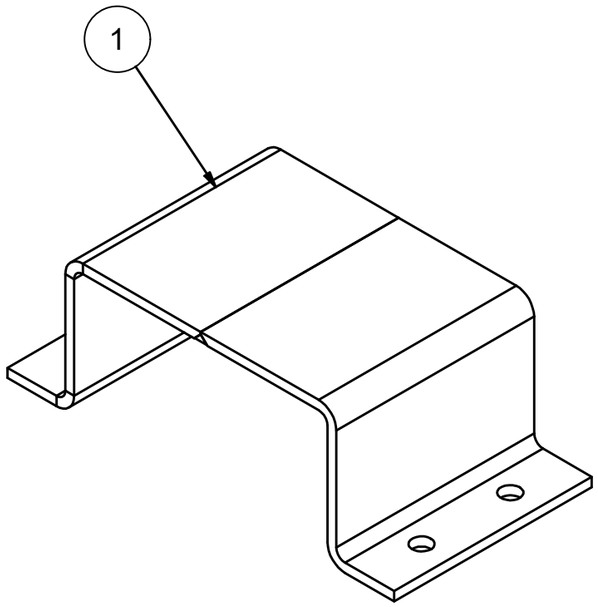
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
3	24989142-04-03-00 JOINT HOUSING BOTTOM RIGHT	1	ALUMINIUM
2	2498142-04-02-00 JOINT HOUSING BOTTOM BACK	1	ALUMINIUM
1	24989142-04-01-00 JOINT HOUSING BOTTON LEFT	1	ALUMINIUM

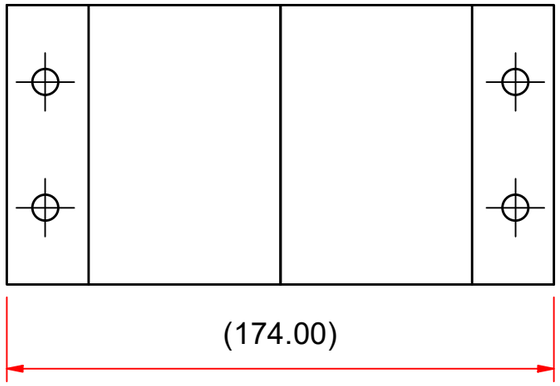
STELLENBOSCH UNIVERSITY

STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A3** = 1:2 **UNITS IN** mm	**TITLE:** JOINT HOUSING BOTTOM	
			DATE 2026/10/01	**SHEET Nr.** OF **SHEETS**	**Nr.** 24989142-04-00-00









UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

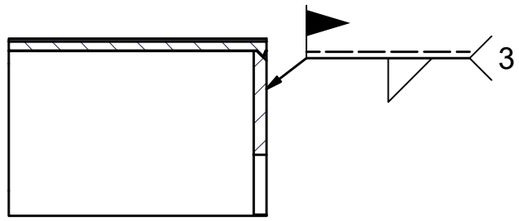
Parts List

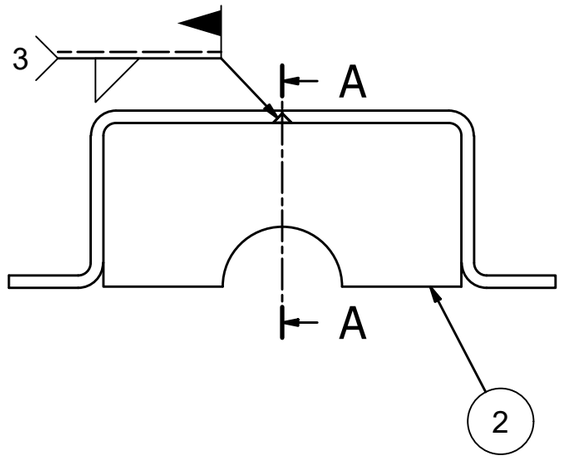
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
1	24989142-05-10-00 JOINT HOUSING TOP HALF	2	ALUMINIUM

STELLENBOSCH UNIVERSITY

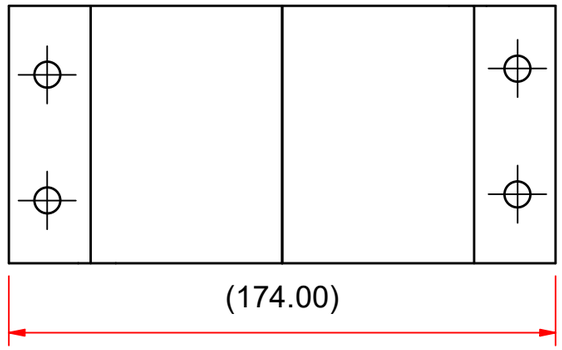
STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**DATE** 2026/10/01	**SCALE ON A3** = 1:2 **UNITS IN** mm	**TITLE:** JOINT HOUSING TOP-FRONT **SHEET Nr.** **OF** **SHEETS** **Nr.** 24989142-05-00-00

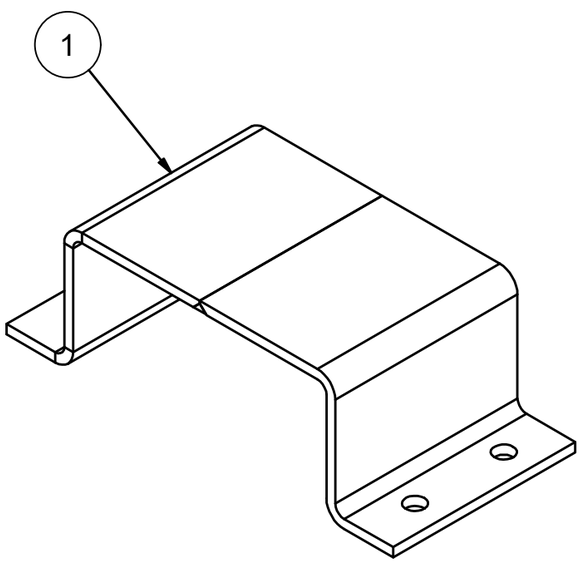
A-A (1 : 2)











UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

Parts List

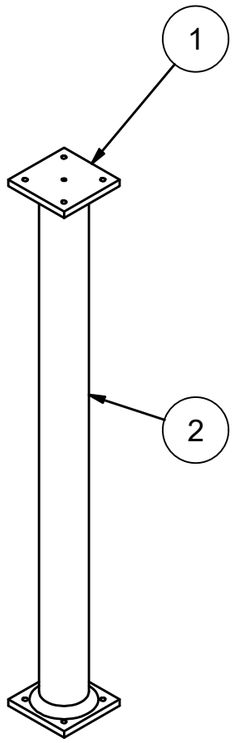
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
2	24989142-04-02-00-00-00 JOINT HOUSING BOTTOM BACK	1	ALUMINIUM
1	24989142-05-01-00-00 JOINT HOUSING TOP FRONT-HALF	2	ALUMINIUM

STELLENBOSCH UNIVERSITY

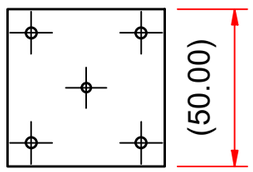
SCALE ON A3 = 1:2
UNITS IN mm

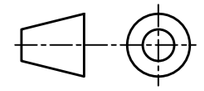
TITLE: JOINT HOUSING TOP-BACK

STUDENT Nr. 24989142 | **DRAWN BY** LN MATOTA | **CHECKED**K SCHREVE | **DATE** 2026/10/01 | **SHEET Nr.** **OF** **SHEETS** | **Nr.** 24989142-06-00-00-00









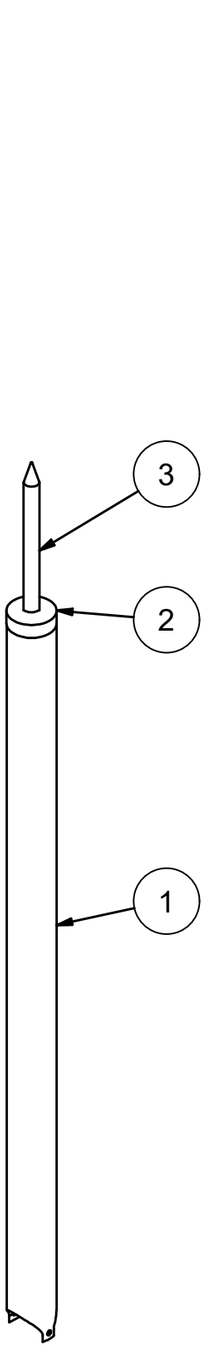
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

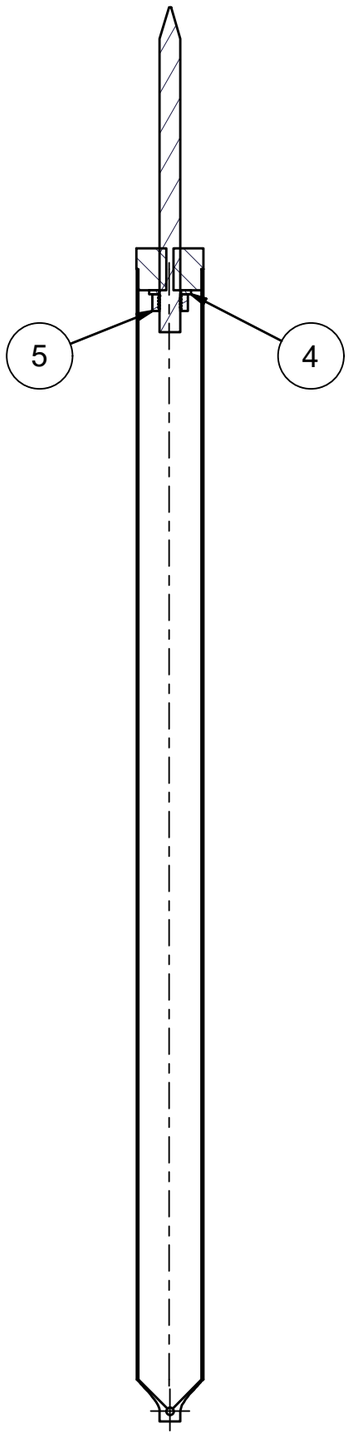
Parts List

ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
2	24989142-07-02-00 BASE LINK PIPE	1	ALUMINIUM
1	24989142-07-01-00 BASE LINK PLATE	2	ALUMINIUM

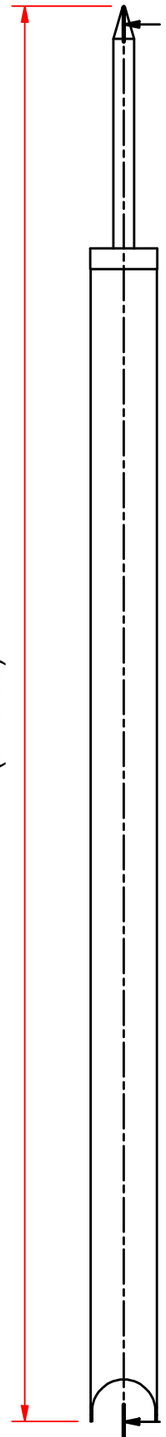
STELLENBOSCH UNIVERSITY

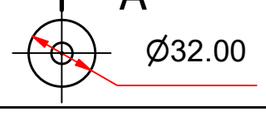
STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**DATE** 2026/10/01	**SCALE ON A**3 = 1:2 **UNITS IN** mm	**TITLE:** BASE LINK **SHEET Nr.** **OF** **SHEETS** **Nr.** 24989142-07-00-00

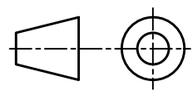




A-A (1 : 3)





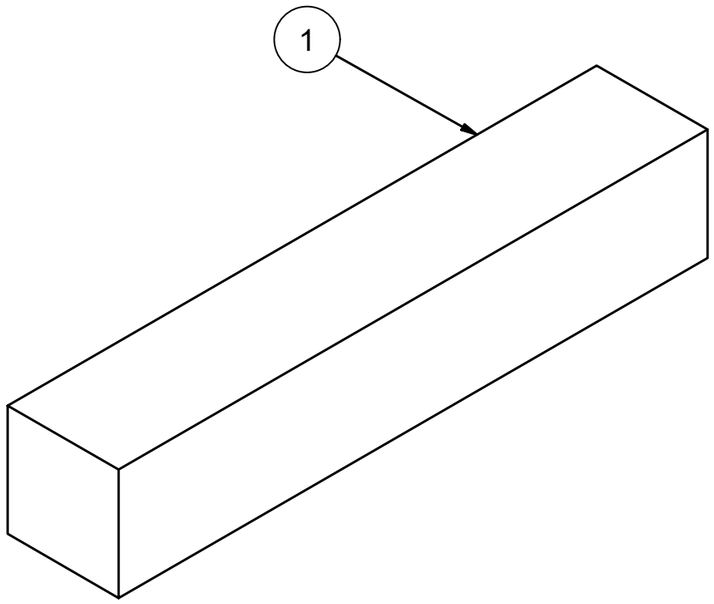


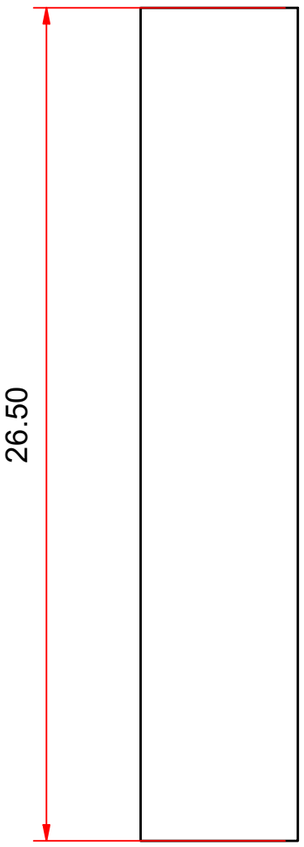
Parts List			
5	M10X1.5 NUT	1	-
4	WASHER ID 10MM	1	-
3	24989142-08-03-00 ARTICULATING LINK PROBE	1	ALUMINIUM
2	24989142-08-02-00 ARTICULATING LINK PROBE HUB	1	ALUMINIUM
1	24989142-08-01-00 ARTICULATING LINK PIPE	1	ALUMINIUM
ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

STELLENBOSCH UNIVERSITY

STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A**3 = 1:3 **UNITS IN** mm	**TITLE:** ARTICULATING LINK	
			DATE 2026/10/01	**SHEET Nr.** **OF** **SHEETS**	**Nr.** 24989142-08-00-00





UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

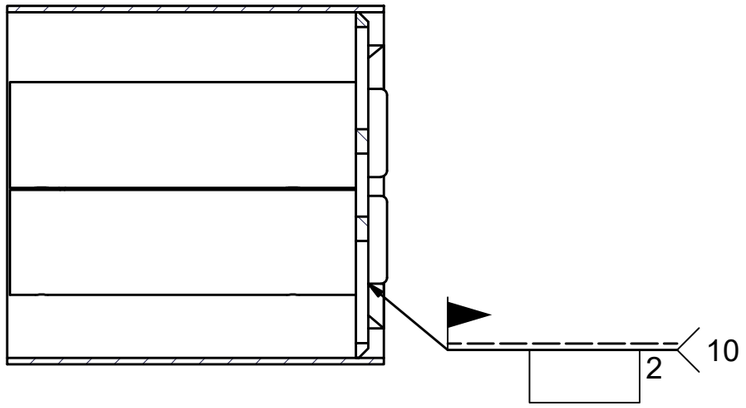
Parts List

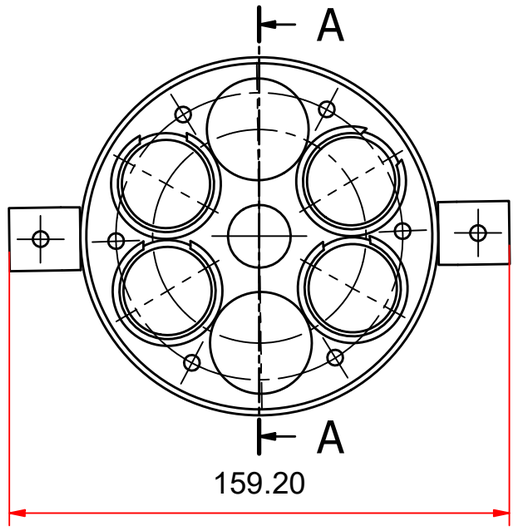
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
1	24989142-09-00-00 BRAKE DISC KEY	1	BRASS

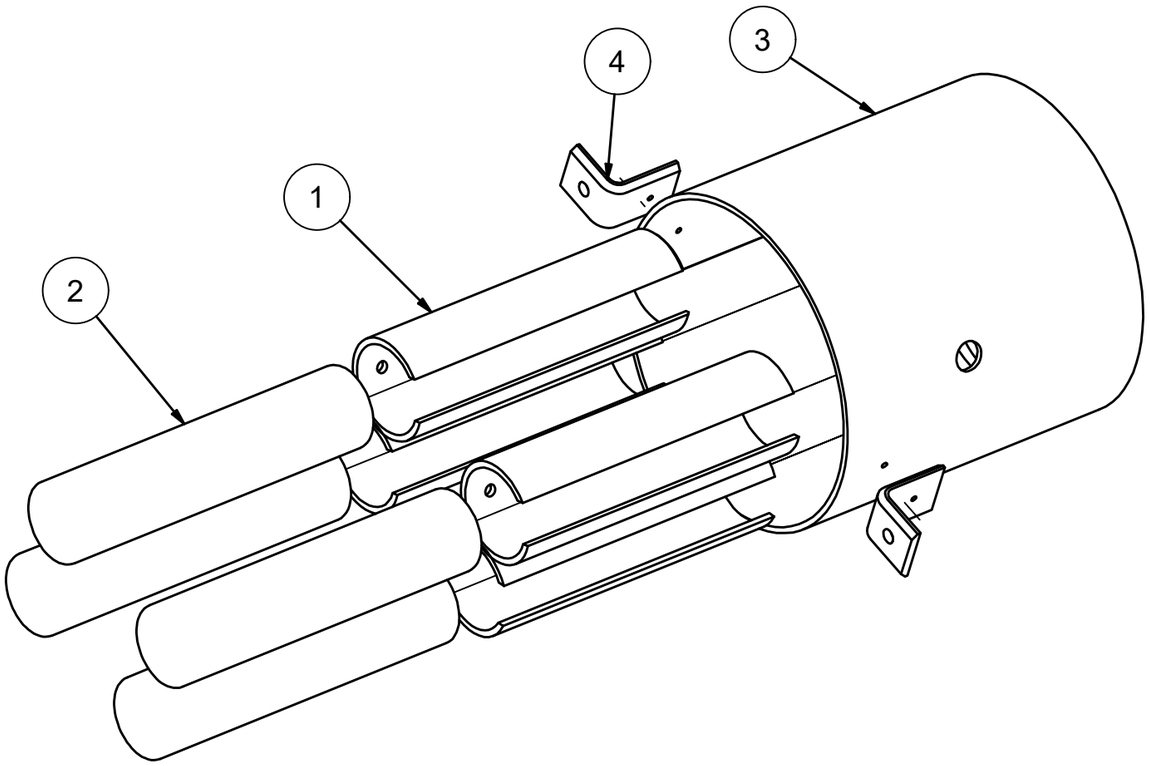
STELLENBOSCH UNIVERSITY

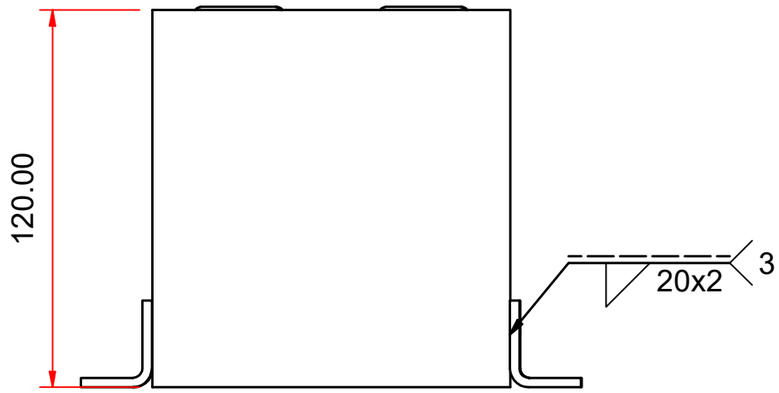
STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**DATE** 2026/10/01	**SCALE ON A**3 = 5:1 **UNITS IN** mm	**TITLE:** BRAKE DISC KEY **SHEET Nr.** **OF** **SHEETS** **Nr.** 24989142-09-00-00

A-A (1 : 2)









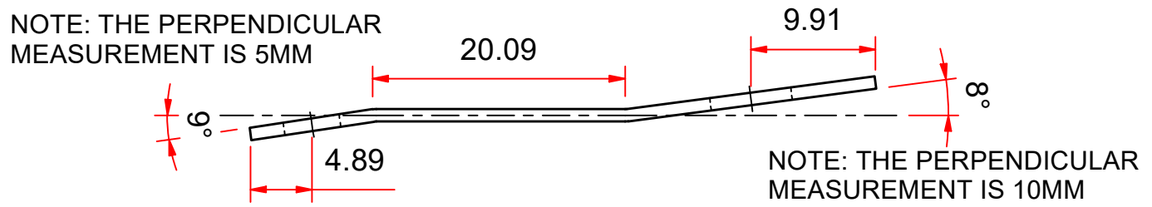
Parts List			
4	24989142-10-04-00 PRESSURE TABS	2	MILD STEEL
3	24989142-10-03-00 ELECTROMAGNETIC COVER	1	MILD STEEL
2	24989142-10-02-00 SOLENOID CORE	4	MILD STEEL
1	24989142-10-01-00 SOLENOID YOKE	4	MILD STEEL
ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

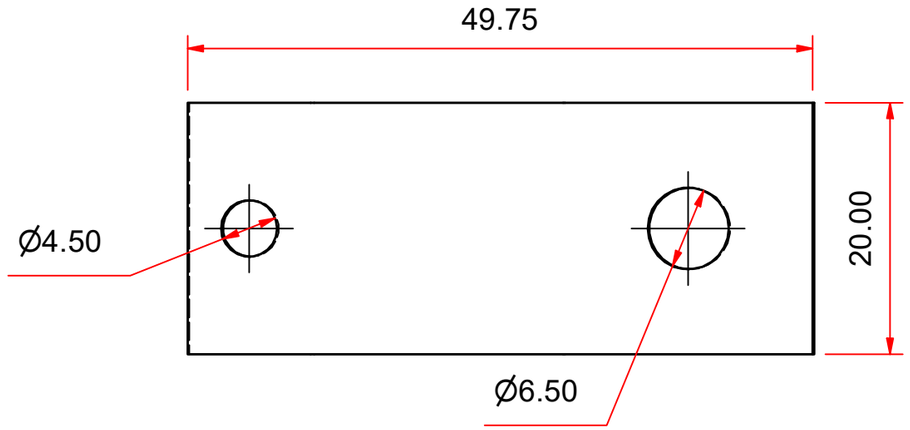
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

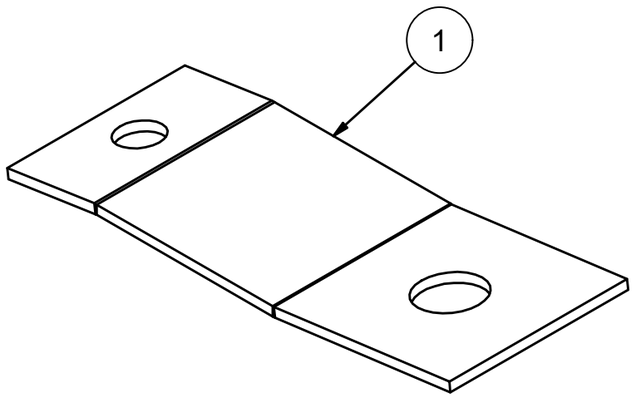
STELLENBOSCH UNIVERSITY

SCALE ON A3 = 1:2
UNITS IN mm
TITLE: ELECTROMAGNET

STUDENT Nr. 24989142 | **DRAWN BY** LN MATOTA | **CHECKED** K SCHREVE | **DATE** 2026/10/01 | **SHEET Nr.** OF **SHEETS** | **Nr.** 24989142-10-00-00







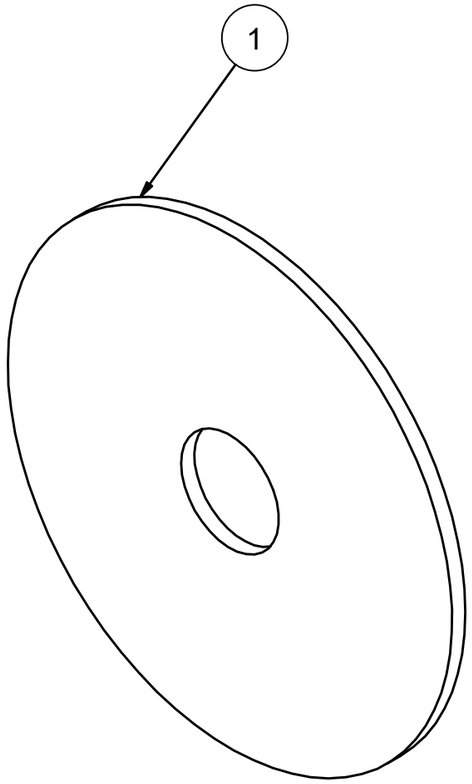
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

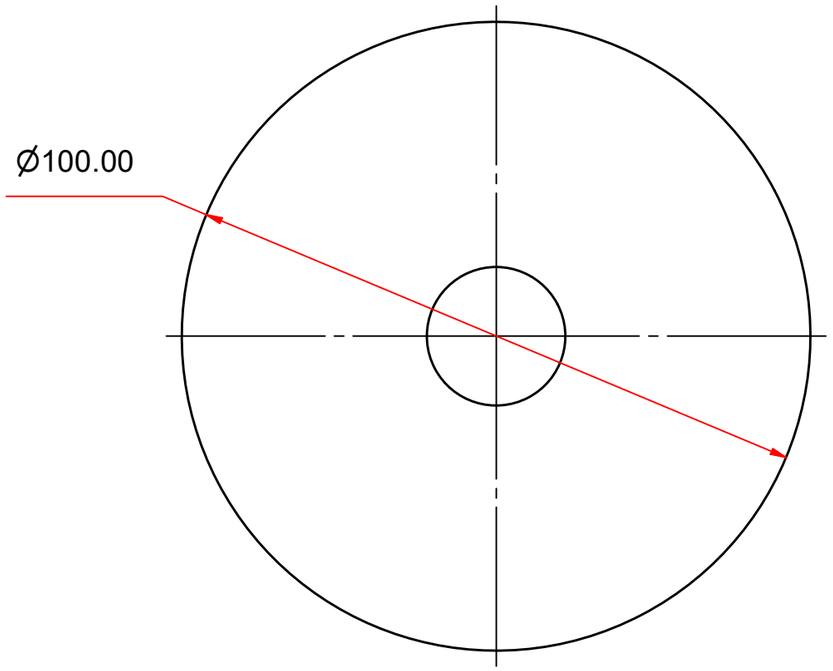
Parts List

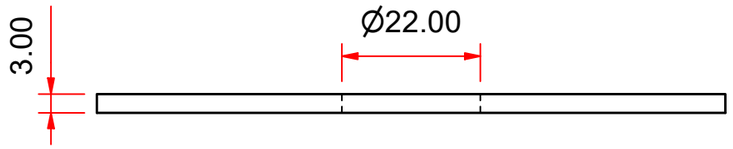
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
1	24989142-11-00-00 PRESSURE TAB	1	MILD STEEL

STELLENBOSCH UNIVERSITY

STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A**3 = 2:1 **UNITS IN** mm	**TITLE:** PRESSURE TAB	**Nr.** 24989142-11-00-00
			DATE 2026/10/01	**SHEET Nr.** OF **SHEETS**	







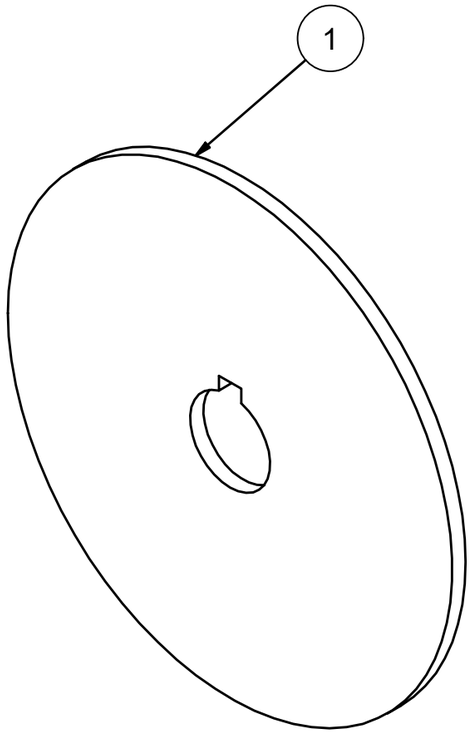
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

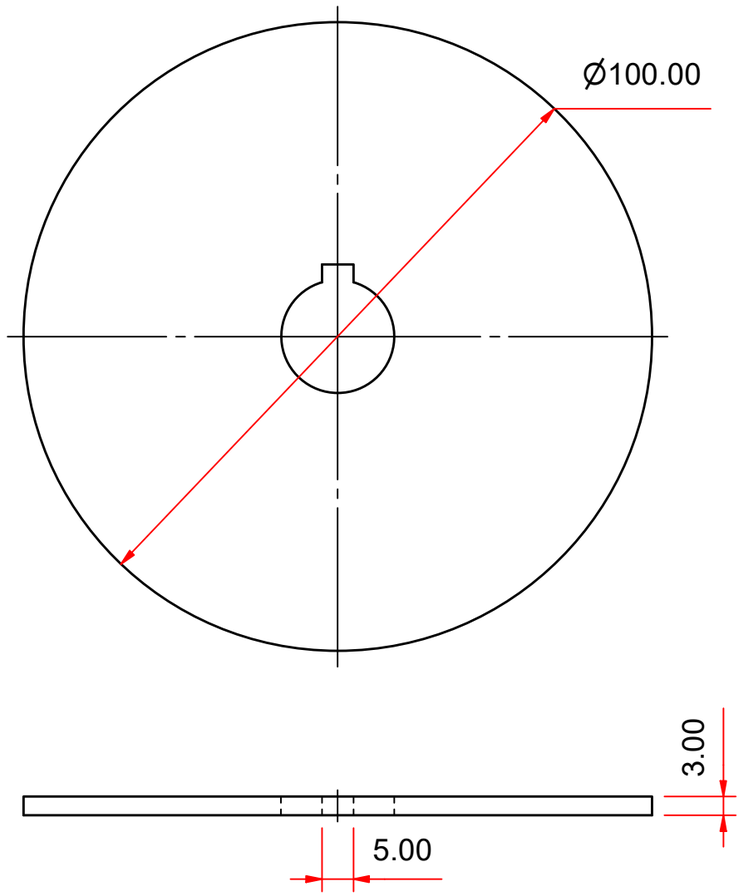
Parts List

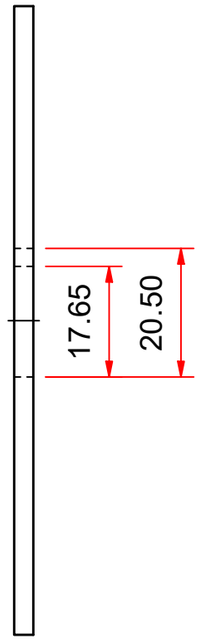
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
1	24989142-14-00-00 FRICTION PLATE	1	MILD STEEL

STELLENBOSCH UNIVERSITY

STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A3** = 1:2 **UNITS IN** mm	**TITLE:** FRICTION PLATE	
			DATE 2026/10/01	**SHEET Nr.** **OF** **SHEETS**	**Nr.** 24989142-14-00-00







UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

Parts List

ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
1	24989142-12-00-00-00 BRAKE DISC	1	WOOD

STELLENBOSCH UNIVERSITY

STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A**3 = 1:1 **UNITS IN** mm	**TITLE:** BRAKE DISC	**Nr.** 24989142-12-00-00-00
			DATE 2026/10/01	**SHEET Nr.** OF **SHEETS**	